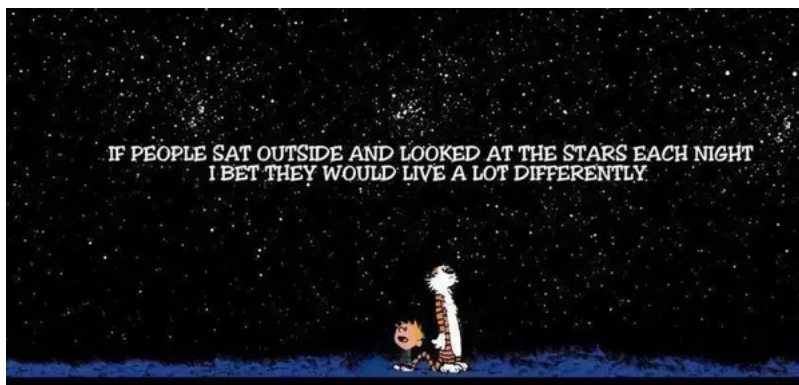


Sailing the Seas of Scientific Notation¹

Your name:

Who you are working with:



Warm-Up

So yesterday we fundamentally changed our understanding of the universe by figuring out that *negative exponents are a thing!* Today we will review the work we did from the last two lessons and work through the some of the questions on the back of the previous sheets.

1. What are the following in exponential form?

(a) $\frac{1}{10^6} \frac{1}{10^{-6}}$

(b) $\frac{10^{-4}}{10^{-14}} \cdot 10^0$

(c) $10000 \cdot 0.0001$

(d) $0.00000001 \cdot 10 \cdot 0.1 \cdot 100000000000$

2. Now let's get some basic scientific notation in here: Scientific notation is always of the form $d.abcd \times 10^n$, so where the first digit (d) is some number 0 through 9, and then there is a decimal point (or comma “,” if you prefer the German method), then there is a series of numbers that could go on for quite a while—the more numbers we have after the decimal point, the more accurate our measurement—but you will see that in Chemistry and other subjects too! So in this form 1234 would be written as 1.234×10^3 . We will get to negative exponents on the next page, but first, try these on for size:

(a) How would you write 7000 or 2300000 or 321321 in this form?

(b) How about 8765432100 or 3141.593?

(c) What is $(2 \times 10^3) \cdot (3 \times 10^2)$ equal to in the form $digit \times 10^{exponent}$?

(d) What about $8 \times 10^8 \cdot 7 \times 10^7$? Check out $6 \times 10^5 \cdot 4 \times 10^3 \cdot 2 \times 10^{-10}$ Remember, your answer should only have one digit to the left of the decimal point!

¹Part 3 of An introduction to Scientific Notation and Scale in five parts, written by Joshua Aaron, Comic by Bill Waterson.

Scientific Notation: A Schmancy² Way of Writing Numbers

The key idea here is that this way of writing numbers not only makes multiplying and dividing these funny looking figures significantly easier but also is *super helpful* when comparing two numbers, and estimating the answers to problems that would otherwise either need a calculator, or just need a lot of raw, opaque calculations:

Shall I Compare Thee to a Summer's Day?³

1. Which is larger 322mm^2 or 2.8cm^2 ?
2. Which is smaller 4.75×10^{-9} liters or 8.18×10^{-12} liters?
3. Which of the following distances would you rather walk?

$$\frac{3.4 \times 10^8}{1.9 \times 10^{12} \cdot 7.1 \times 10^{-5}} \text{ kilometers} \quad \text{or} \quad 1.2 \times 10^{10} \cdot \frac{2.2 \times 10^{-3}}{5.5 \times 10^4} \text{ meters}$$

Using Your Brain⁴

Okay, so here is the part where I pose some questions that are a bit trickier and you answer them (first trying individually of course, then working in pairs or small groups). These are questions that in some way involve estimation—that is, the answers aren't really exact, they are more “highly educated guesses.” So the important thing here is *how you get your answers* and that your answer *makes sense*, it is not really a problem if you and your friend next to you have long decimals that look different. ***Be sure to answer these on a separate sheet of paper!*** With those ideas in mind, let's get started!

1. How many smartphones (in their original boxes) can fit in a shipping container? To answer this question try the following:
 - (a) What is your best guess for the dimensions of a normal smartphone box?
 - (b) What is the volume of this box?
 - (c) A standard 20foot shipping container that travels on the biggest container ships (but are also suitable for rail or truck transport) have interior dimensions of approximately $5.9\text{m} \times 2.4\text{m} \times 2.4\text{m}$. What is it's volume?
 - (d) Is the best method to just divide the above numbers, or is there another, perhaps more realistic, but also somewhat simpler way? (You might want to try the division first, and come back to think about other methods...**Hint** think about making a giant shape out of little toy blocks!)
2. Approximately many such containers would we need to ship 12 million smartphones?
3. Suppose we find a way of making the smartphone box using 25% less volume than you decided on for question 1, approximately how many shipping containers would we now need to ship the same number of phones? If we divide this answer by our previous one, what percent do we get?
4. In the previous problems we used the interior measurements of a standard shipping container. Given that the largest container ships these days carry over 21000 such containers, how many cubic meters of freight can such a behemoth⁵ carry?
5. Suppose all of that cargo has a density around that of water (1 gram per cubic centimeter), how many tonnes would a full container ship have? (1 metric ton is 1000 kg.)

²This is a technical term

³From a sonnet by Willy Shakes!

⁴If you forgot your brain today, please let me know, my Brother is in town so he may have brought me an extra from Canada.

⁵Behemoth means “really big hulk of a thing” but is based on the name for a mythical creature—worth researching at home!